

From-Scratch Mini LLM: Pre-training and Sentiment Transfer

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Abstract

This report describes a compact, from-scratch decoder-only language model inspired by the Qwen3 family. The model (19.52M parameters) uses Qwen3-family design choices (RMSNorm, rotary position embeddings, SwiGLU, and grouped-query capable attention) at a size that trains on CPU; it is not a reproduction of Qwen at scale. It is pre-trained on the TinyStories corpus with a causal language-modeling objective and then reused for a binary sentiment task. No pre-trained weights are used at any stage. Pre-training reaches a final validation perplexity near 13.9, and the frozen representations support a sentiment probe at 64.19% accuracy and 64.03% macro-F1, above a 56.00% majority-class floor. Fine-tuning the base for the same task raises accuracy to 83.00%, showing that the representations hold substantially more sentiment signal than a frozen probe can access. A tokenizer coverage analysis further shows that although out-of-vocabulary (OOV) rates on the sentiment domain are high (20.42%), per-sentence OOV does not predict misclassification, so adaptation rather than vocabulary coverage is the dominant lever. The work demonstrates the full path from architectural components to a working transfer-learning result on commodity CPU hardware.

1 Introduction

Modern large language models combine a small set of architectural ideas: rotary position embeddings [1], RMSNorm [2], and gated feed-forward layers [3], all built on the transformer attention mechanism [4] and popularized at scale by model families such as Qwen [5]. This project implements those pieces directly and at a size that trains on CPU, using the TinyStories dataset [6] for pre-training. The learned representations are then evaluated on a downstream sentiment task built from an emotion-labeled sentence corpus [7].

2 Model architecture

The model is a decoder-only transformer with the following choices: pre-normalization with RMSNorm, rotary position embeddings applied to queries and keys, SwiGLU feed-forward blocks, and tied input/output embeddings. Multi-head attention supports grouped-query attention, though the reported run uses standard multi-head attention. Table 1 lists the configuration.

3 Data and tokenization

Pre-training uses the first 10,000 TinyStories, split 80/10/10 into training, validation, and test sets with a fixed seed. A word-level tokenizer is trained on the training split only, so held-out text does not influence the vocabulary. The tokenizer keeps four special tokens (padding, beginning of text, end of text, and unknown) and selects the most frequent remaining tokens, yielding a 7,392-token

Table 1: Model configuration.

Hyperparameter	Value
Parameters	19,520,000
Hidden size (d_{model})	512
Layers	6
Attention heads	8
FFN intermediate size	1,024
Maximum sequence length	256
Vocabulary size	7,392
Tied embeddings	Yes

vocabulary that covers 99.84% of training-token occurrences. Sequence-length analysis (Figure 1) informed the choice of a 256-token maximum, which balances coverage against the memory cost of longer contexts on CPU.

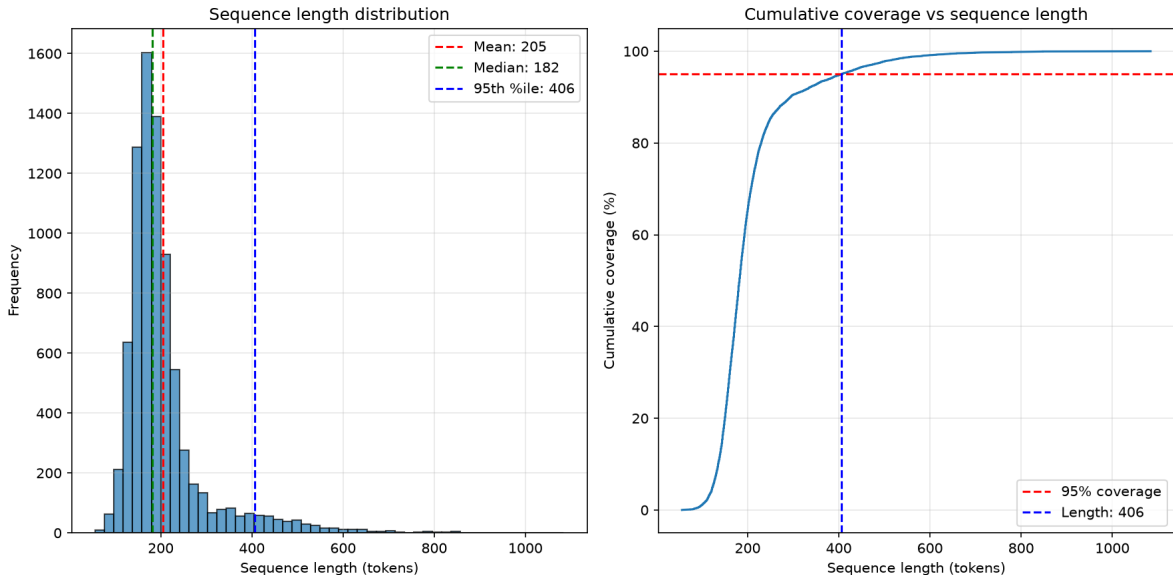


Figure 1: Token-length distribution of the training stories and cumulative coverage as a function of sequence length.

4 Pre-training

The model is trained for five epochs (5,000 optimizer steps) with a causal language-modeling loss. Optimization uses AdamW [8] with $\beta_2 = 0.95$, which is the Qwen setting and is more stable than the usual 0.999 on short sequences, together with gradient clipping and a cosine learning-rate schedule with linear warmup. Table 2 summarizes the outcome, and Figure 2 shows the loss and perplexity curves.

Table 2: Pre-training results.

Metric	Value
Total steps	5,000
Final training loss / perplexity	1.84 / 6.31
Final validation loss / perplexity	2.63 / 13.90
Best validation loss	2.61

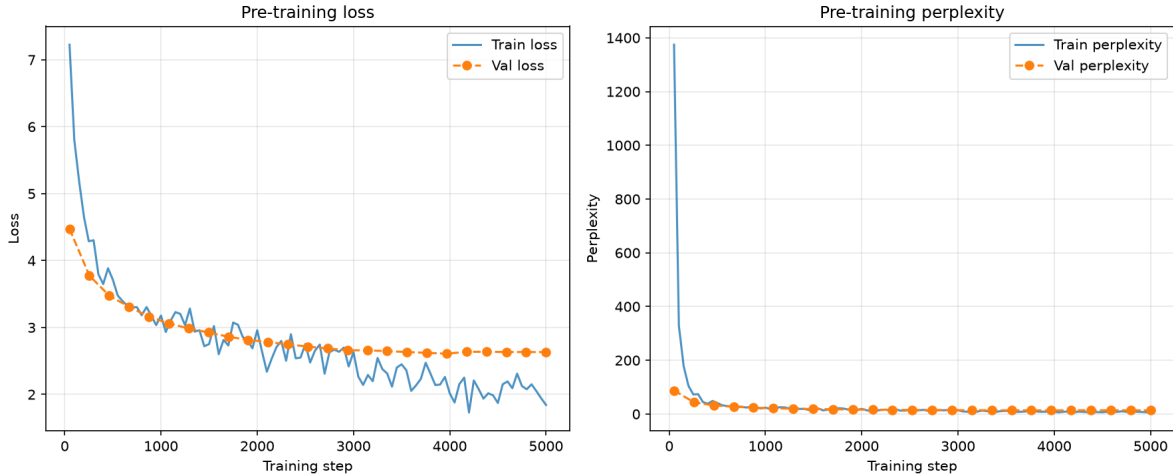


Figure 2: Pre-training loss (left) and perplexity (right) over training steps, for both the training batches and periodic validation checkpoints.

5 Generation

Text generation supports greedy decoding as well as temperature, top- k , and nucleus (top- p) sampling, with optional repetition and no-repeat- n -gram penalties. A key-value cache makes single-prompt generation incremental; rotary positions are offset by the cached length so a cached decode matches a full forward pass token for token. At this model scale the samples are coherent within a few sentences and reflect the simple TinyStories style rather than long-range narrative structure.

6 Sentiment transfer

The downstream task is binary sentiment classification over 16,000 emotion-labeled sentences (7,238 positive, 8,762 negative), split 80/10/10 with the same fixed seed, giving a 1,600-sentence test set. The pre-trained model is frozen and its hidden states are mean-pooled into a single vector per sentence, which feeds a small MLP head (one 128-unit hidden layer). Only the head is trained, for 15 epochs at learning rate 10^{-3} . Freezing the base turns the task into a probe: it measures how much sentiment signal the language model captured during pre-training while keeping training inexpensive. Table 3 reports test-set metrics with 95% percentile-bootstrap confidence intervals (1,000 resamples over the fixed test set), and Figures 3 and 4 show the confusion matrix and per-class breakdown.

Table 3: Frozen-probe sentiment results (test set). Confidence intervals are 95% percentile bootstrap over the test set.

Metric	Value	95% CI
Accuracy	64.19%	[61.81, 66.63]
Macro-F1	64.03%	[61.64, 66.55]
Negative precision	69.96%	[66.79, 73.12]
Negative recall	63.17%	[60.02, 66.34]
Negative F1	66.39%	[63.84, 68.95]
Positive precision	58.28%	[54.66, 61.95]
Positive recall	65.48%	[61.96, 68.91]
Positive F1	61.67%	[58.73, 64.78]
Best validation macro-F1	66.69%	

7 Baselines and comparison

A probe score is only interpretable against reference points. We compare the frozen probe to a majority-class floor, a lexical baseline (TF-IDF features with logistic regression), and the identical model with its base fine-tuned rather than frozen. All four use the same test split. Table 4 and Figure 5 summarize the comparison.

Table 4: Sentiment methods on the shared test set.

Method	Accuracy	Macro-F1
Majority class	56.00%	35.90%
Frozen probe (ours)	64.19%	64.03%
Fine-tuned base (ours)	83.00%	82.56%
TF-IDF + logistic regression	93.19%	93.04%

The frozen probe clears the majority floor with non-overlapping confidence intervals, so the pre-trained features carry genuine sentiment signal. Unfreezing the base lifts accuracy from 64.19% to 83.00% (macro-F1 64.03% to 82.56%), recovering most of the distance to the lexical baseline. The frozen probe therefore understates the information in the representations; adaptation, not feature quality, is the dominant lever.

8 Tokenizer coverage and error analysis

Because the tokenizer was fit on TinyStories, it covers only 79.58% of sentiment tokens (20.42% OOV), and a natural hypothesis is that OOV caps the probe. The data does not support this at the sentence level. The correlation between a sentence’s OOV rate and being misclassified is -0.06 ; mean OOV is essentially equal for correct (22.78%) and incorrect (21.50%) predictions; and, as Figure 6 shows, accuracy does not decline as OOV rises. OOV is a real indicator of the domain gap but not the binding constraint here, which is consistent with fine-tuning rather than vocabulary delivering the large gain.

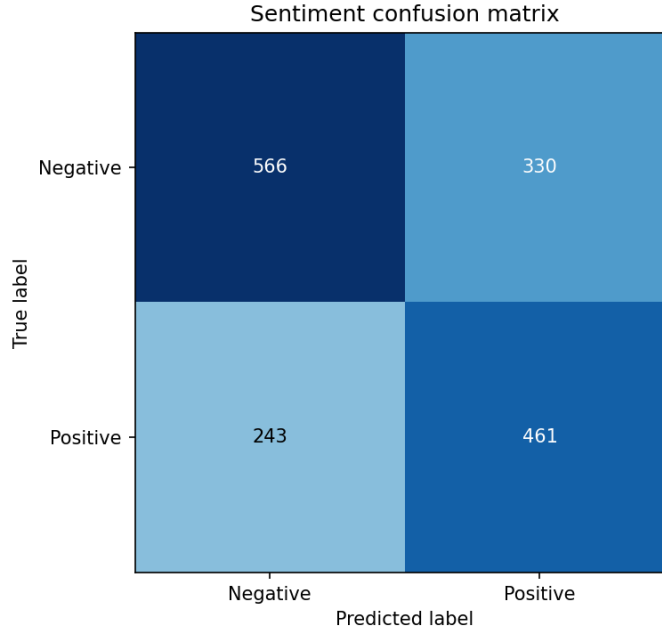


Figure 3: Sentiment confusion matrix on the test set (rows are true labels, columns are predictions).

9 Discussion and conclusion

A 19.52M-parameter model trained on 10,000 short stories learns representations that transfer to sentiment classification well above the majority floor, despite the large domain gap between children’s stories and emotion-labeled sentences. The frozen probe favors the negative class slightly, consistent with the class balance in the sentiment data, and lands at 64.19% accuracy. The more informative result is comparative: fine-tuning the base reaches 83.00%, and the OOV analysis rules out vocabulary coverage as the sentence-level bottleneck, so representation adaptation is the highest-impact lever. The absolute numbers are modest but expected given the model size, training budget, and CPU-only setting; the emphasis here is on a correct, transparent implementation of every stage with honest baselines and uncertainty rather than on maximizing a benchmark. Natural extensions include subword tokenization, a larger model and training budget on GPU, and confidence intervals over repeated training runs to capture seed variance in addition to test-set sampling.

References

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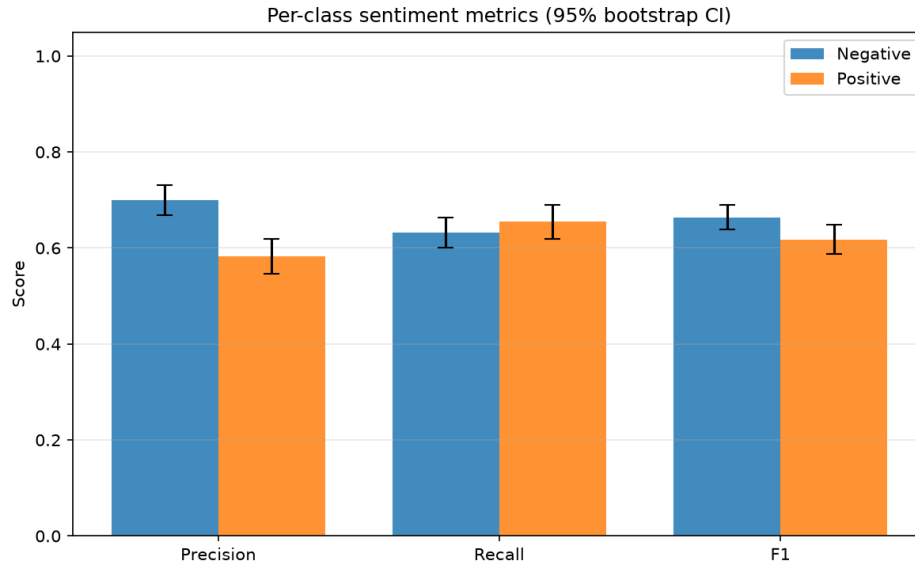


Figure 4: Per-class precision, recall, and F1 for the frozen probe, with 95% bootstrap confidence intervals as error bars.

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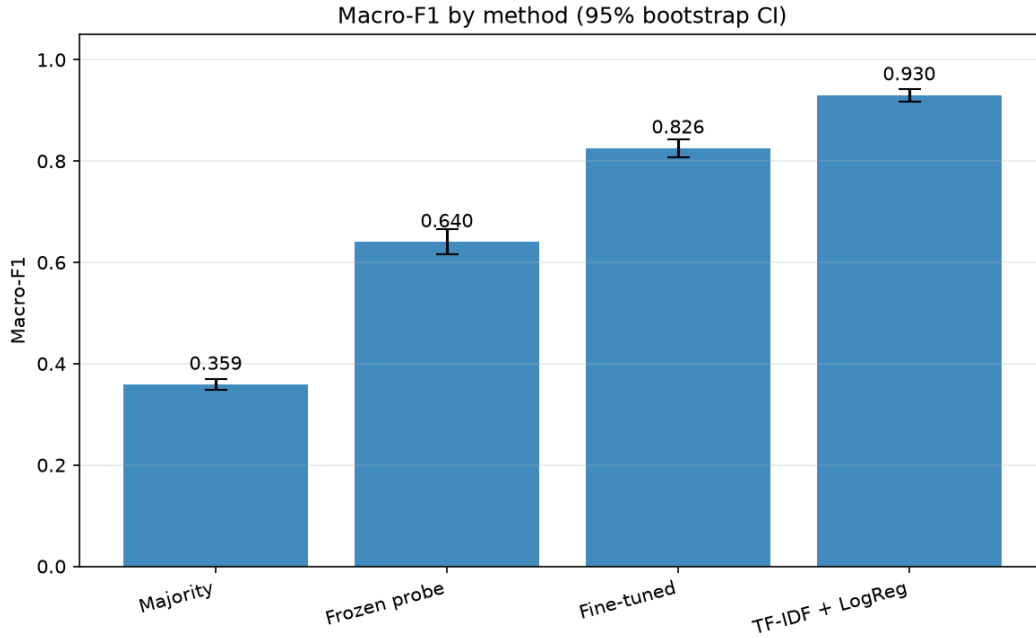


Figure 5: Macro-F1 by method with 95% bootstrap confidence intervals. The frozen probe clears the majority floor with non-overlapping intervals; fine-tuning closes most of the gap to the lexical baseline.

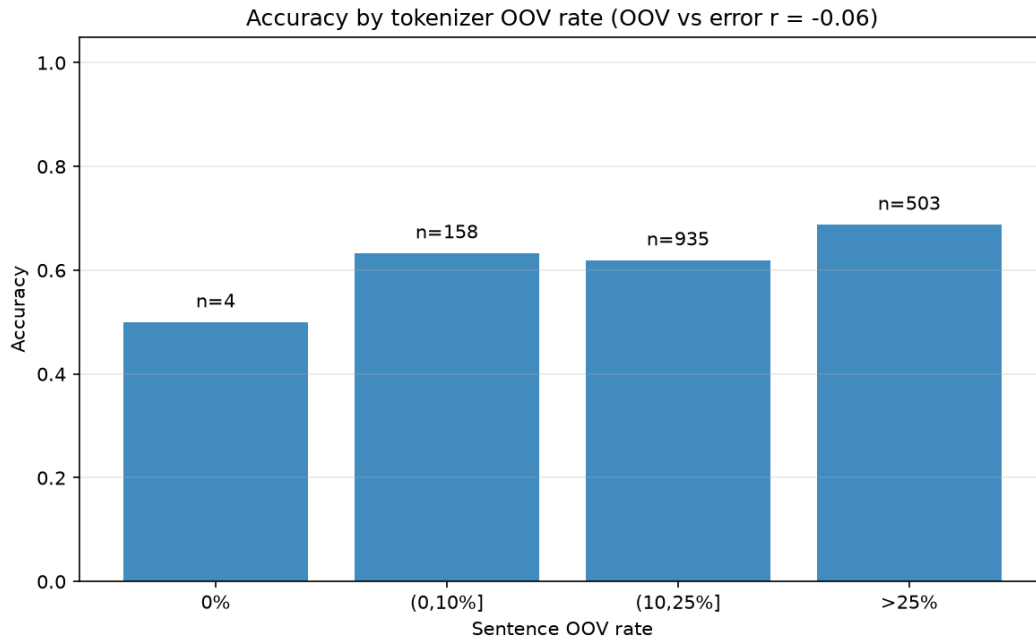


Figure 6: Test accuracy by per-sentence OOV-rate bucket. Accuracy is roughly flat (and even highest in the largest-OOV bucket), so sentence-level OOV does not predict error.